

TB Detection System – Project Specification

■ 1. Machine Learning Model Design

- Framework: TensorFlow/Keras (preferred for deployment flexibility and transfer learning).
- Model Type: Fine-tuned CNN using DenseNet121 pretrained on ImageNet.
- Dataset: Kaggle's 'Tuberculosis (TB) Chest X-ray Database'.
- Model Hosting: Separate FastAPI microservice hosted on Render (free tier).
- Storage for Future Retraining: Yes, predictions and X-ray data stored in database.

■ 2. Application Functionality

- Patients can upload multiple X-rays over time to track progress.
- AI predicts TB severity (Mild, Moderate, Severe).
- Doctor review/approval system: AI predicts → Doctor confirms → Report finalized.
- AI Explainability: Grad-CAM heatmaps generated and attached to reports.
- Reports: Downloadable as PDF and viewable online via unique QR code.

■ 3. User Management & Roles

- Roles: Patient, Doctor, Admin.
- Admin adds doctors manually.
- Patients can sign up freely.
- 2-Step Email Verification (OTP sent to Gmail).
- Doctors upload manual signatures for digital reports.

■ 4. Data Handling & Storage

- Storage: Supabase (preferred free option with large storage).
- Stored Data: Raw image + AI heatmap output.
- Sensitive Data Encryption in MongoDB Atlas.
- Audit Logs maintained for every upload and prediction.

■ 5. Security & Privacy

- 2FA login enabled for all roles.
- Role-Based Access Control: Doctors and Admins have higher privileges.
- Anonymization of uploaded image metadata before analysis.
- Basic HIPAA/GDPR-inspired data privacy implementation.

■ 6. Analytics & Insights

- Charts: Monthly TB-positive case trends, AI confidence averages, doctor-verified accuracy.
- Geo-Mapping of TB case hotspots by city.
- Admins can export analytics as CSV or PDF.
- Patients can view public model performance stats and reliability scores.

■ 7. Notifications & Communication

- Email notifications for report readiness and doctor reviews.
- Weekly summary reports emailed to doctors (case statistics).
- Integrated chatbot using Gemini API for TB awareness and FAQs.
- Feedback system for user experience and app improvement.

■ 8. DevOps & Deployment

- Frontend: Vercel.
- Backend: Vercel (Node.js + Express + JWT Authentication).
- ML API: Render (FastAPI-based microservice).
- Dockerized architecture with scalable containers.
- CI/CD pipeline with automatic deployments on code commits.

■ 9. Research & Reporting

- Used for both academic research and public demonstration.
- Performance metrics logged: Accuracy, Precision, Recall, F1-score.
- Research Mode: Logs every new AI prediction for retraining analysis.
- Feedback loop integrated – doctors' confirmations retrain future versions.

■ Additional Features

- User feedback section to review experience and suggest improvements.
- Automatic periodic accuracy reports for Admin review.
- Heatmap overlay comparison for before/after X-rays of same patient.
- Comprehensive dashboard combining AI and doctor analytics.